Answer key

Practice Test 4 (State level) · 50 questions · 34 with checked app keys

Each answer lists short steps, the vault note (Question Types folder) and the app keys. Every key line was run in the app and its screen value matched the computed answer (check_keys.py). Accept numeric answers within about 2% with the right unit.

Multiple-choice answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14
D	A	B	B	C	A	B	C	A	A	B	B	A	B

15	16	17
B	A	B

Section I: Multiple choice

1. D. A thermocouple (copper and bismuth, one junction hot, one cold)

1. Poggendorff suggested a thermoelectric source (Seebeck effect, 1821).
2. Ohm's thermocouple: copper and bismuth, one junction in boiling water, one in ice.
3. Its voltage stayed steady, so he could measure current against wire length.

Note: A1 History/Ohm
App: no app: fact or concept

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2. A. Zinc and copper (or silver)

1. Each cell was zinc + copper (or silver) with a brine-soaked separator.
2. Stacking cells in series added their voltages.

Note: A1 History/Volta
App: no app: fact or concept

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3. B. Claude Shannon

1. Shannon's MIT master's thesis, "A Symbolic Analysis of Relay and Switching Circuits" (1937).
2. It joined Boole's 1854 algebra to real switching circuits.

Note: A1 History/Boole
App: no app: fact or concept

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4. B. The elementary charge e

1. 2019 SI: e = $1.602176634 \times 10^{-19}$ C exactly.
2. One ampere is then a set number of elementary charges per second.

Note: A1 History/Units Named After People
App: no app: fact or concept

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5. C. The Teflon is negative and the wool is positive

1. The material near the negative end grabs electrons.
2. Teflon gains electrons (negative); wool loses them (positive).

Note: A2 Static Electricity and Coulomb/Charge Basics
App: no app: fact or concept

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6. A. Zero

1. Each charge makes a field of the same size at the midpoint.
2. The two fields point in opposite directions, so they cancel.

Note: A2 Static Electricity and Coulomb/Electric Field and Potential
App: no app: fact or concept

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7. B. Their internal resistance is far too high to give hundreds of amps

1. A starter motor needs about 100-200 A.
2. A coin cell has an internal resistance of many ohms.
3. Most of the EMF is lost inside the cells, so the current stays small.

Note: A3 DC and AC/Batteries and Cells
App: no app: fact or concept

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8. C. 100 mA

1. About 1 mA: you just feel it. About 10 mA: you may not be able to let go.
2. About 100 mA through the chest: ventricular fibrillation is likely.

Note: A3 DC and AC/Shock and Safety
App: no app: fact or concept

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9. A. 3.60 MJ

1. 1 kWh = 1000 W × 3600 s.
2. = 3 600 000 J = 3.60 MJ.

Note: A4 Current Voltage Resistance Power Energy/Power Energy and Cost
App: `J p1000 h1` → `j Energy (joules) = 3.6 MJ` *(checked)*

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10. A. 9.0 V

1. No current flows (the circuit is open), so there is no voltage drop across R1.
2. The meter sees the full battery voltage across the gap: 9.0 V.
3. This is how you find an open part: the full supply appears across it.

Note: A5 DC Circuits/Open and Short Circuits
App: no app: fact or concept

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11. B. 3

1. SPDT = single pole, double throw.
2. One common (COM), one normally open (NO), one normally closed (NC).
3. 3 contact terminals.

Note: A6 Switches Relays Fuses Breakers GFCI/Relays
App: no app: fact or concept

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12. B. It senses arcing, such as from a damaged cord or a loose connection

1. AFCI: detects the current pattern of an electric arc (a fire risk).
2. A 5 mA leak is the GFCI's job. Surges are handled by surge protectors.

Note: A6 Switches Relays Fuses Breakers GFCI/Circuit Breakers
App: no app: fact or concept

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13. A. A small current is sent around the sensor through a resistor; the imbalance makes it trip

1. The test button sends a few mA from hot back to neutral by a path that skips the sensor.
2. The hot and neutral currents no longer match, so the GFCI trips like a real ground fault.

Note: A6 Switches Relays Fuses Breakers GFCI/GFCI
App: no app: fact or concept

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14. B. Reverse biased, in parallel with the load

1. A zener holds a near-constant voltage in reverse breakdown.
2. Placed across (parallel with) the load, it holds the load voltage steady.
3. A series resistor takes up the extra supply voltage.

Note: A7 Diodes and LEDs/Special Diodes
App: no app: fact or concept

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15. B. 2

1. The select lines must name 4 inputs.
2. 2 bits give $2^2 = 4$ choices.

Note: A8 Digital Logic/Multiplexers and Decoders
App: no app: fact or concept

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16. A. A′B

1. (X + Y)' = X'Y'.
2. (A + B')' = A' · B'' = A'B.

Note: A8 Digital Logic/De Morgan and Bubble Pushing
App: `t-(a+-b)` → `F = A'B` *(checked)*

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17. B. The meter is almost a short circuit across the battery; a large current flows and the meter fuse may blow

1. The 10 A jack goes through a tiny shunt resistance inside the meter.
2. Across a battery, that is nearly a short: a big current flows.
3. Always move the red lead back to the V jack after a current measurement.

Note: A9 Hands-On and Meters/Multimeter Use
App: no app: fact or concept

Section II: Short answer and calculations

18. 1.5 N, attractive

1. $F = k|q_1 q_2|/r^2$, with r = 0.18 m.
2. $F = 8.99e9 \times 1.2e\text{-}6 \times 4.5e\text{-}6 / 0.18^2 = 1.4983$ N = 1.5 N (2 significant figures).
3. Opposite signs attract.

Note: A2 Static Electricity and Coulomb/Coulombs Law
App: `q a1.2u b-4.5u r.18` → `f Force (+ repel) = -1.498 N` *(checked)*

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19. 2.0 × 10^{11} electrons

1. n = |Q| / e.
2. n = 32 × 10^{-9} / 1.602 × 10^{-19} = 1.998e+11.
3. 2 significant figures: 2.0 × 10^{11} electrons.

Note: A2 Static Electricity and Coulomb/Charge Basics
App: `a q32n` → `e Electrons = 1.9973E11 electrons` *(checked)*

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20. x = 0.20 m

1. Like charges: the fields cancel BETWEEN them, closer to the smaller charge.
2. k(2.0)/x^2 = k(8.0)/(0.60 – x)^2, so (0.60 – x)/x = √4 = 2.
3. 0.60 = 3x, so x = 0.20 m.

Note: A2 Static Electricity and Coulomb/Electric Field and Potential
App: `Q 2u 0, 8u .6` → `E = 0 at x = 0.2 m` *(checked)*

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21. 12 V

1. V_{peak} = 34 / 2 = 17 V.
2. V_{rms} = 17 / √2 = 12.02 V = 12 V.

Note: A3 DC and AC/AC Characteristics
App: `w a34` → `r RMS = 12.02 V` *(checked)*

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22. 10.8 V; 17 min (0.28 h)

1. Series cells add voltage: 3 × 3.6 = 10.8 V (about 11 V). The capacity stays 3400 mAh.
2. t = 3.4 Ah / 12 A = 0.2833 h.
3. × 60 = 17.0 min, about 17 min.

Note: A3 DC and AC/Batteries and Cells
App: `p s3 v3.6 c3400` → `a Pack voltage = 10.8 V` *(checked)*
App: `b c3400 i12` → `t Time (hours) = 0.28333 h` *(checked)*

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23. 0.675 Ω

1. A = πr^2 = π(0.750 × 10^{-3})^2 = 1.7671e-06 m^2.
2. R = ρL/A = 2.65e-8 × 45.0 / 1.7671e-06 = 0.675 Ω.

Note: A4 Current Voltage Resistance Power Energy/Resistance and Resistivity
App: `r p2.65E-8 l45 d1.5m` → `r Resistance = 0.67482 ohm` *(checked)*

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24. $31 (223.2 kWh; $31.25 before rounding)

1. Energy = 1.20 kW × 6.0 h × 31 = 223.2 kWh.
2. Cost = 223.2 × $0.14 = $31.25.

Note: A4 Current Voltage Resistance Power Energy/Power Energy and Cost
App: `e p1200 h6 d31 r.14` → `c Cost = $31.25 (31.248)` *(checked)*

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25. 1.80 kJ

1. P = V^2/R = 144 / 24.0 = 6.00 W.
2. t = 5.00 × 60 = 300. s.
3. E = Pt = 6.00 × 300. = 1.80 kJ.

Note: A4 Current Voltage Resistance Power Energy/Power Energy and Cost
App: `J v12 r24 t5min` → `j Energy (joules) = 1.8 kJ` *(checked)*

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26. 5.60 kΩ, ±2%

1. Digits: green 5, blue 6, black 0 = 560.
2. Multiplier: brown = ×10, so 5600 ohm.
3. Tolerance: red = ±2%.

Note: A5 DC Circuits/Resistor Color Codes
App: `c gbknr` → `= 5600 ohm` *(checked)*
App: `c gbknr` → `5.6 kohm +/-2%` *(checked)*

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27. 683 = 68 kΩ; 2R2 = 2.2 Ω

1. 3-digit code: two digits, then the number of zeros: 68 + 000 = 68 000 ohm = 68 kohm.
2. R marks the decimal point: 2R2 = 2.2 ohm.

Note: A5 DC Circuits/Resistor Color Codes
App: `c683` → `If SMD code: 683 = 68 kohm` *(checked)*
App: `c2R2` → `SMD code 2R2 = 2.2 ohm` *(checked)*

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28. 140 Ω

1. 100 || 150 = 100 × 150 / 250 = 60 ohm.
2. 47 + 60 + 33 = 140 ohm.

Note: A5 DC Circuits/Series and Parallel Resistance
App: `s47+100/150+33` → `Req = 140 ohm` *(checked)*

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29. 0.100 Ω

1. At full scale the movement has 2.00 mA × 50.0 ohm = 0.100 V across it.
2. The shunt carries the rest: 1.00 A − 0.00200 A = 0.998 A.
3. R_{sh} = 0.100 V / 0.998 A = 0.100 Ω.

Note: A5 DC Circuits/Meters and Measurement
App: `o v2m*50 i1-2m` → `r Resistance = 0.1002 ohm` *(checked)*

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30. 36.0 mA

1. Current divider: $I_{330} = I \times R_{other} / (R_1 + R_2)$.
2. = 90.0 mA × 220 / 550 = 36.0 mA.
3. Check: the bigger resistor takes the smaller share.

Note: A5 DC Circuits/Series and Parallel Resistance
App: `k t90m a220 b330` → `z I2 (in R2) = 36 mA` *(checked)*

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31. 14 A; a 15 A fuse

1. Parallel loads: powers add: 1650 W.
2. I = 1650 / 120 = 13.75 A.
3. 10 A is too small; the next size, 15 A, carries 14 A.

Note: A6 Switches Relays Fuses Breakers GFCI/Fuses
App: `u p1000+650 v120` → `i Current = 13.75 A` *(checked)*
App: `u p1000+650 v120` → `Smallest std fuse above I: 15 A` *(checked)*

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32. 1.2×10^2 Ω (120 Ω); 48 mW

1. LED drops add in series: 3 × 3.2 = 9.6 V.
2. Resistor voltage = 12.0 – 9.6 = 2.4 V (2 significant figures).
3. R = 2.4 / 0.020 = 120 ohm. P = 2.4 × 0.020 = 48 mW.

Note: A7 Diodes and LEDs/LEDs
App: `l s12 f3.2 c3 i20m` → `r Resistor = 120 ohm` *(checked)*
App: `l s12 f3.2 c3 i20m` → `p Resistor power = 48 mW` *(checked)*

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33. F = A′C′ + BC

1. Minterms 000, 010, 011, 111.
2. 0 and 2 (A = 0, C = 0): A'C'.
3. 3 and 7 (B = 1, C = 1): BC.
4. A'B (2 and 3) is not needed: both are already covered.

Note: A8 Digital Logic/Boolean Algebra and Simplification
App: `t abc m0,2,3,7` → `F = A'C' + BC` *(checked)*

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34. F = B

1. Rows 0, 1, 4, 5 are 000, 001, 100, 101: exactly the rows with B = 0.
2. So F = 1 whenever B = 1: F = B.

Note: A8 Digital Logic/Truth Tables and Boolean Expressions
App: `t abc M0,1,4,5` → `F = B` *(checked)*

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35. 10010101 = 149

1. 1100111 = 103, 101110 = 46.
2. 103 + 46 = 149.
3. 149 = 128 + 16 + 4 + 1 = 10010101.

Note: A8 Digital Logic/Binary Numbers and Addition
App: `x1100111+101110` → `sum bin 10010101` *(checked)*

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36. F = AB + AC′

1. F = A · (B + C').
2. Distribute: F = AB + AC'.
3. Truth table check: F = 1 for ABC = 100, 110, 111.

Note: A8 Digital Logic/Logic Design Word Problems
App: `ta(b+-c)` → `F = AB + AC'` *(checked)*

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37. 1.37 kΩ

1. Series: the same current flows in both.
2. I = 4.62 / 2200 = 2.100 mA.
3. R_x = 2.88 V / I = 2200 × 2.88 / 4.62 = 1.37 kΩ.

Note: A9 Hands-On and Meters/Task - Mystery Resistor
App: `m a2.2k v4.62 w2.88` → `x Mystery R (Rx) = 1.371 kohm` *(checked)*

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38. (a) 15 W, 18 W, 18 W; (b) 1.1 h (about 68 min)

1. P = VI: 5 × 3 = 15 W; 9 × 2 = 18 W; 12 × 1.5 = 18 W.
2. Battery energy = 4.5 Ah × 3.87 V = 17.4 Wh.
3. Useful charging power = 18 W × 0.85 = 15.3 W.
4. t = 17.4 / 15.3 = 1.1 h.

Note: A9 Hands-On and Meters/Task - USB Charger
App: `z c4500 v3.87 s9 a2 e.85` → `t Charge time = 1.1382 h` *(checked)*

Section III: Circuit problems

Problem A: two batteries (Kirchhoff's laws) (Questions 39–42)

E1 = 18.0 V and E2 = 15.0 V are ideal batteries (the long plate, +, is at the top of each). R1 = 2.00 Ω, R2 = 4.00 Ω, R3 = 6.00 Ω. Let I_1 be the current in R1 flowing toward node A and I_2 the current in R2 flowing toward node A.

39. 8.00 I_1 + 6.00 I_2 = 18.0 and 6.00 I_1 + 10.0 I_2 = 15.0

1. KCL: the current down through R3 is $I_1 + I_2$.
2. Left loop: 18.0 = 2.00 I_1 + 6.00($I_1 + I_2$).
3. Right loop: 15.0 = 4.00 I_2 + 6.00($I_1 + I_2$).
4. Collect the terms: 8 I_1 + 6 I_2 = 18 and 6 I_1 + 10 I_2 = 15.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: no app: fact or concept

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40. 2.05 A

1. Determinant: 8 × 10 − 6 × 6 = 44.
2. I_1 = (18 × 10 − 6 × 15) / 44 = 90/44 = 2.05 A.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `j a8 b6 c18 d6 e10 f15` → `x x (I1 or V1) = 2.0455` *(checked)*
App: `N V1–0 18 1–2 2 V3–0 15 3–2 4 2–0 6` → `V1–0 18: 2.045 A out of +, gives 36.82 W` *(checked)*

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41. 0.273 A toward A; E2 gives power (it is discharging)

1. I_2 = (8 × 15 − 6 × 18) / 44 = 12/44 = 0.273 A.
2. I_2 is positive: the current leaves the + plate of E2 and flows toward A.
3. So E2 is discharging (giving power).

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `j a8 b6 c18 d6 e10 f15` → `y y (I2 or V2) = 0.27273` *(checked)*
App: `N V1–0 18 1–2 2 V3–0 15 3–2 4 2–0 6` → `V3–0 15: 272.7 mA out of +, gives 4.091 W` *(checked)*

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42. V_{AB} = 13.9 V; P_{R3} = 32.2 W

1. $I_3 = I_1 + I_2 = 2.0455 + 0.2727 = 2.3182$ A.
2. $V_{AB} = 6.00 \times 2.3182 = 13.9$ V.
3. $P = I^2R = 2.3182^2 \times 6.00 = 32.2$ W.
4. Node check: $(V - 18)/2 + (V - 15)/4 + V/6 = 0$ gives $V = 153/11 = 13.9$ V.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 18 1–2 2 V3–0 15 3–2 4 2–0 6` → `V1 = 18 V V2 = 13.91 V` *(checked)*
App: `N V1–0 18 1–2 2 V3–0 15 3–2 4 2–0 6` → `R2–0 6: 13.91 V 2.318 A 2>0 32.24 W` *(checked)*

Problem B: ladder network (Questions 43–45)

A 24.0 V ideal battery feeds the ladder below. The series resistors R1, R3 and R5 are each 1.00 kΩ. R2 and R4 are each 2.00 kΩ. R6 is 1.00 kΩ. X, Y and Z are the top nodes.

43. 2.00 kΩ

1. Start at the far end: R5 + R6 = 2.00 k.
2. That is in parallel with R4: 2k || 2k = 1.00 k.
3. Add R3: 2.00 k. In parallel with R2: 1.00 k.
4. Add R1: R_{eq} = 2.00 kΩ.

Note: A5 DC Circuits/Hard Resistor Networks
App: `N 1–2 1k 2–0 2k 2–3 1k 3–0 2k 3–4 1k 4–0 1k =1–0` → `R(1–0) = 2 kohm` *(checked)*
App: `s1k+2k/(1k+2k/(1k+1k))` → `Req = 2 kohm` *(checked)*

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44. 6.00 mA

1. Battery current = 24.0 / 2.00 k = 12.0 mA.
2. $V_X = 24.0 - 12.0$ mA $\times$ 1.00 k = 12.00 V.
3. I_{R2} = 12.00 V / 2.00 k = 6.00 mA.

Note: A5 DC Circuits/Hard Resistor Networks
App: `N V1–0 24 1–2 1k 2–0 2k 2–3 1k 3–0 2k 3–4 1k 4–0 1k` → `R2–0 2k: 12 V 6 mA 2>0 72 mW` *(checked)*

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45. 3.00 V

1. Each step halves the voltage: V_X = 12.00 V.
2. Y: R3 (1k) in series with 1.00 k (R4 || (R5 + R6)): V_Y = 6.00 V.
3. Z: R5 and R6 are equal: $V_Z = 6.00/2 = 3.00$ V.

Note: A5 DC Circuits/Dividers and Bulb Brightness
App: `N V1–0 24 1–2 1k 2–0 2k 2–3 1k 3–0 2k 3–4 1k 4–0 1k` → `V3 = 6 V V4 = 3 V` *(checked)*

Problem C: logic circuit (Questions 46–47)

An XOR gate and a NOR gate are connected as shown. The inputs are A, B, C; the output is F.

46. F = ((A ⊕ B) + C)′ = A′B′C′ + ABC′

1. F = NOR(A XOR B, C) = ((A ⊕ B) + C)'.
2. De Morgan: F = (A ⊕ B)' · C' = (A XNOR B) C'.
3. A XNOR B = A'B' + AB, so F = A'B'C' + ABC'.

Note: A8 Digital Logic/De Morgan and Bubble Pushing
App: `t-((a^b)+c)` → `F = A'B'C' + ABC'` *(checked)*

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47. 000 and 110: m(0, 6)

1. F = 1 needs C = 0 and A = B.
2. ABC = 000 (m0) and 110 (m6).

Note: A8 Digital Logic/Truth Tables and Boolean Expressions
App: `t-((a^b)+c)` → `m(0,6)` *(checked)*

Problem D: diodes in parallel branches (Questions 48–50)

A 9.00 V ideal battery feeds node X through R1 = 1.00 kΩ. From X to the bottom wire there are three branches: a silicon diode D1 (0.70 V when on) with R2 = 1.00 kΩ; a red LED (2.00 V when on) with R3 = 1.00 kΩ; and a silicon diode D3 with its anode on the bottom wire.

48. D1 and the LED conduct; D3 is off (reverse biased)

1. X is above the bottom wire, so D1 and the LED are forward biased.
2. X (about 3.9 V) is at least 2.0 V, so the LED also turns on.
3. D3 has its cathode at X and its anode at 0 V: reverse biased, off.

Note: A7 Diodes and LEDs/Diode Circuit Analysis
App: `N V1-0 9, 1-2 1k, D2-3, 3-0 1k, D2-4 2, 4-0 1k, D0-2` → `D0-2 .7: OFF, -3.9 V across` *(checked)*

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49. 3.90 V

1. KCL at X: (9.00 − V)/1k = (V − 0.70)/1k + (V − 2.00)/1k.
2. 9.00 − V = 2V − 2.70, so 3V = 11.70.
3. V_X = 3.90 V.

Note: A7 Diodes and LEDs/Diode Circuit Analysis
App: `N V1-0 9, 1-2 1k, D2-3, 3-0 1k, D2-4 2, 4-0 1k, D0-2` → `V1 = 9 V V2 = 3.9 V` *(checked)*

---

50. 1.90 mA

1. I = (V_X − 2.00)/1.00 k = (3.90 − 2.00)/1000 = 1.90 mA.
2. Check: D1 carries 3.20 mA; total 5.10 mA = (9.00 − 3.90)/1k.

Note: A7 Diodes and LEDs/Diode Circuit Analysis
App: `N V1-0 9, 1-2 1k, D2-3, 3-0 1k, D2-4 2, 4-0 1k, D0-2` → `D2-4 2: ON 1.9 mA 2>4 3.8 mW` *(checked)*
